Could HSAT2 Repeat RNAs Drive Long COVID and ME/CFS? A New Hypothesis

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Human satellite 2 (HSAT2) is an abundant class of repetitive DNA sequences that normally remain silent under physiological conditions. In cancer, however, HSAT2 may become aberrantly expressed and reprogram cells, ultimately promoting immune suppression. Here we propose that an analogous mechanism may contribute to the development of post-acute infection syndromes (PAIS), including Long COVID and ME/CFS.

Human satellite 2 (HSAT2) is a pericentromeric repeat representing approximately 1.5% of the human genome (Fig. 1) [1]. Under physiological conditions, it is transcriptionally silenced through DNA methylation and packaging into a compact chromatin state known as heterochromatin [2-5] (Fig. 2).

In cancer, however, HSAT2 may become aberrantly expressed [6-8]. The resulting HSAT2 RNAs can reprogram gene expression through molecular mechanisms that are increasingly well understood [5,8] (Fig. 2-4). Remarkably, this reprogramming is not restricted to the cells producing HSAT2 RNAs, as these RNAs are released into body fluids within extracellular vesicles (EVs), possibly allowing their dissemination to distant tissues [7-9].

In cancer, it was recognized early on that tumors expressing HSAT2 exhibit a "cold tumor" phenotype, characterized by a profoundly immunosuppressive microenvironment. Subsequent studies have provided mechanistic support for this observation by showing that HSAT2 RNAs impair several layers of the body's defense mechanisms [7,8,10] (Fig. 2). These include innate immunity, which allows virtually every cell to defend itself against viral infection, and the immune system, whose specialized cells not only protect the body against pathogens but also coordinate communication between organs and promote tissue repair.

Here we propose that an analogous mechanism, involving HSAT2 RNAs as a central player and possibly other related satellite RNAs, such as HSAT3 [1,8], may contribute to the development of post-acute infection syndromes (PAIS), including Long COVID and ME/CFS.

Our attention was drawn to this possibility by the observation that **HSAT2 RNAs may be induced upon herpesvirus infection or reactivation**, as shown for HCMV [11]. Replication of viruses belonging to the herpesvirus family disrupts centromere organization [12-15], and this is likely the mechanism by which they can induce transcription of adjacent pericentromeric satellite repeats, including HSAT2 (Fig. 3). **This may therefore provide an explanation for the frequent observation that Long COVID and ME/CFS emerge following reactivation of latent herpesviruses**.

Because each member of the herpesvirus family preferentially infects only one or a limited number of cell types, the initial production of RNAs from these repeats may occur in different cellular compartments depending on the virus involved. For example, among herpesviruses reported to reactivate in Long COVID, HSV-1 establishes latency primarily in sensory neurons, EBV in memory B cells, HHV-6 in cells of the central nervous system and the immune system, and HCMV in cells of the myeloid lineage, endothelial cells lining blood vessels, and other tissue-resident cells. Interestingly, the sensory ganglia in which HSV-1 establishes latency are located in close anatomical proximity to the brainstem, a region that has repeatedly been reported to exhibit inflammation in Long COVID. Reactivation of latent HSV-1 could therefore be promoted by this inflammatory environment. In addition, both HCMV and SARS-CoV-2 can infect endothelial cells, raising the possibility of synergistic effects within this cellular compartment.

Following repeated or closely spaced viral infections, cumulative changes in chromatin composition and DNA methylation of HSAT2 repeats may become sufficiently extensive to trigger significant epigenetic erosion, ultimately establishing a permissive chromatin state for HSAT2 repeat transcription. Once initiated, this state could become self-sustaining through continued HSAT2 RNA production, possibly in cooperation with additional mechanisms involving multiple viruses and multiple cell types.
Depending on the magnitude of the initial epigenetic disruption, this transition from transient HSAT2 activation to sustained expression could also occur following a single episode of viral reactivation.

By analogy with cancer, we hypothesize that extracellular vesicles released by cells in which latent herpesviruses have reactivated carry HSAT2 RNAs together with other regulatory RNAs, such as microRNAs, that can reprogram recipient cells. These vesicles could spread from cell to cell and potentially throughout the body, behaving in some respects like an infectious agent by propagating this pathogenic signal and thereby driving persistent immune dysregulation, particularly immune exhaustion (Fig. 4).

Rather than presenting definitive proof, this short Perspective assembles the available evidence into a coherent mechanistic framework supporting the "HSAT2 hypothesis" for Long COVID and ME/CFS, while generating specific, experimentally testable predictions that can now be investigated using patient-derived samples.
It also brings together several key observations reported in Long COVID and ME/CFS that have so far remained difficult to reconcile, including herpesvirus reactivation, immune exhaustion, and persistent systemic dysfunction.

Patients living with Long COVID and ME/CFS share the same goal: recovery.
The fact that some patients recover without apparent long-term sequelae strongly suggests that these disorders are, at least in part, functional rather than irreversible, and are maintained by self-reinforcing pathological feedback loops, possibly involving multiple interacting mechanisms, rather than permanent tissue destruction. We propose that HSAT2 RNAs, both at their initial sites of production and possibly after dissemination throughout the body via EVs, may represent one such mechanism, contributing to both disease initiation and long-term maintenance.
At present, this remains a hypothesis. However, if correct, it would open an entirely new therapeutic perspective: neutralizing extracellular HSAT2 RNAs [7] could interrupt one or more pathological feedback loops, allowing physiological regulatory networks to re-establish themselves and thereby promoting recovery.

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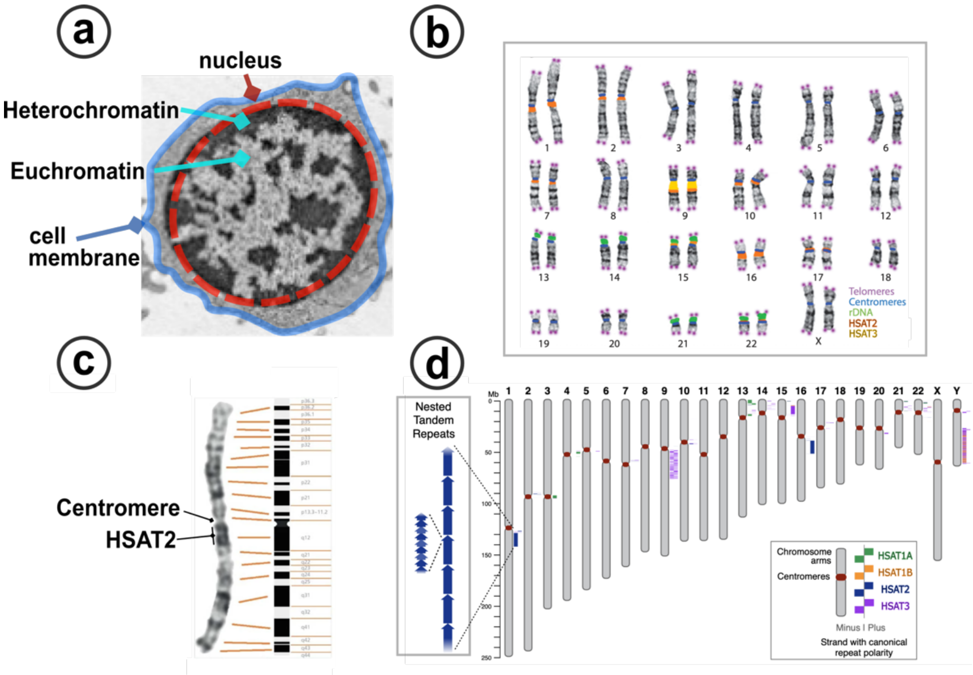

Figure 1. Euchromatin, **heterochromatin**, **satellite repeat sequences**. **(a)** Chromatin in the cell nucleus (shown here in a B lymphocyte) is organized into two major compartments: euchromatin, which contains most genes and where genes can be active, and heterochromatin**,** which appears darker and contains fewer genes, most of which are inactive. **(b)** During cell division, chromosomes become individually visible, revealing the 23 pairs of human chromosomes **(**karyotype**)** with their characteristic pattern of dark and light Giemsa bands [5]. The dark bands correspond to heterochromatin. Chromosome 1 is enlarged in **(c).** Near the central constriction (centromere), all chromosomes contain a more or less extended dark region composed of tandemly repeated DNA sequences, illustrated schematically in the inset in (d) (nested tandem repeats). **(d)** Four major families of pericentromeric satellite repeats (HSAT1A, HSAT1B, HSAT2, and HSAT3) are present on different human chromosomes [1]. HSAT2 and HSAT3 are both built around the repeated DNA motif GGAAT. The location of HSAT2 is highlighted in orange in (b).

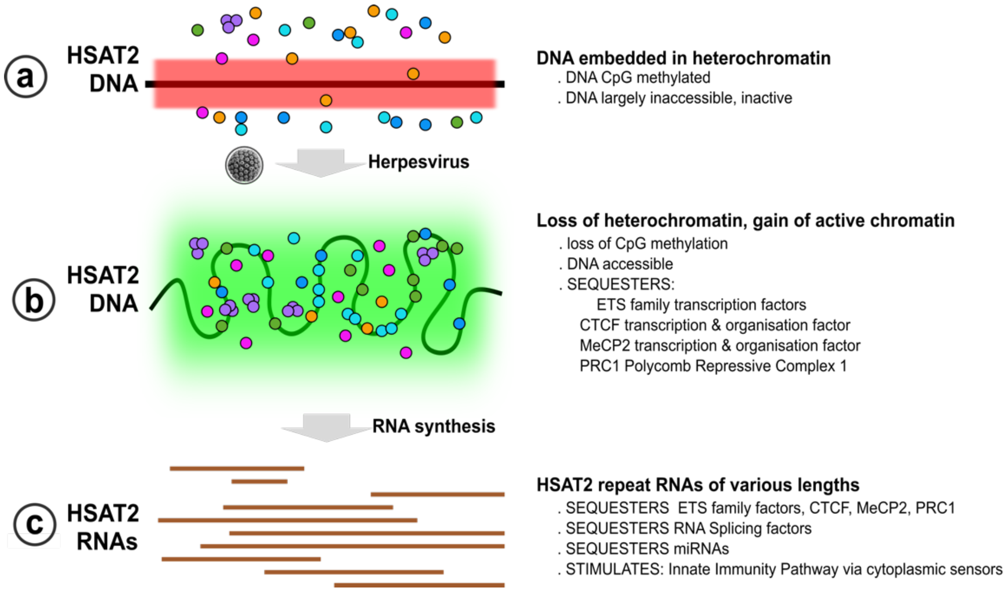

Figure 2. Unfolding of HSAT2 arrays and production of HSAT2 RNAs can reprogram gene expression. **(a)** Under normal conditions, HSAT2 arrays are embedded within heterochromatin and remain largely inaccessible. **(b)** During herpesvirus infection or reactivation, and in many cancers, disruption of heterochromatin unfolds these large DNA arrays, exposing thousands of repeated DNA sequences containing numerous protein-binding motifs, including the GGAA motif recognized by ETS-family transcription factors [5]. These repeated sequences recruit a defined set of proteins that normally control which genes are turned on or off. In particular, they recruit members of the ETS family of transcription factors (including FLI1, PU.1, ETS1, ETS2 and ERG). These proteins both activate HSAT2 transcription and become sequestered on the repetitive DNA instead of binding the genes they normally control elsewhere in the genome. Other proteins that help control gene activity in a more global manner, including CTCF, PRC1 and MeCP2, may also become sequestered. PU.1 and MeCP2 deserve special consideration. PU.1 plays a central role in the development of immune cells, suggesting that its sequestration could broadly impair immune immune cell development and function [17]. MeCP2 plays a central role in the organization of chromatin in neurons and is also particularly important in stem cells, suggesting that these cell types may be especially vulnerable to HSAT2 activation [18]. **(c)** Some of these factors also drive the production of abundant HSAT2 RNAs of different lengths. These RNAs may in turn sequester additional proteins involved in gene activity, including the same or additional transcription factors, RNA splicing factors and microRNAs, while also activating innate immunity pathways present in every cell of the body. Together, these DNA- and RNA-mediated mechanisms may simultaneously alter the activity of many genes. Their consequences are expected to differ from one tissue to another, and may also be influenced by infection with other viruses, such as SARS-CoV-2, providing a framework by which HSAT2 activation could induce widespread cellular reprogramming.

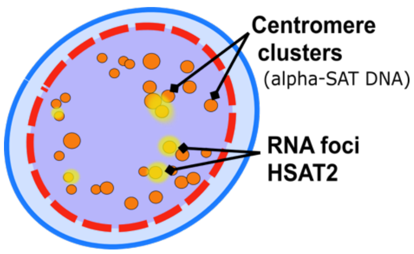

Figure 3. Schematic representation of a cell infected by a herpesvirus. A probe which marks all centromeres reveals their characteristic arrangement into clusters (orange). During herpesvirus infection or reactivation, viral proteins profoundly alter the behavior of centromeres. Several chromosomes, particularly chromosomes 1 and 16, which contain the largest HSAT2 arrays, begin to produce large amounts of HSAT2 RNA [11]. Sites of active HSAT2 RNA production appear as RNA foci, shown here in yellow. Notably, other satellite repeats, such as HSAT3, as well as interspersed repeat sequences **(**transposable elements**)**, may also be induced, as has been observed in other contexts, including a wide range of cancers. This schematic integrates observations from several published studies and is intended as a conceptual illustration rather than a representation of a single microscopy experiment [11,15,16].

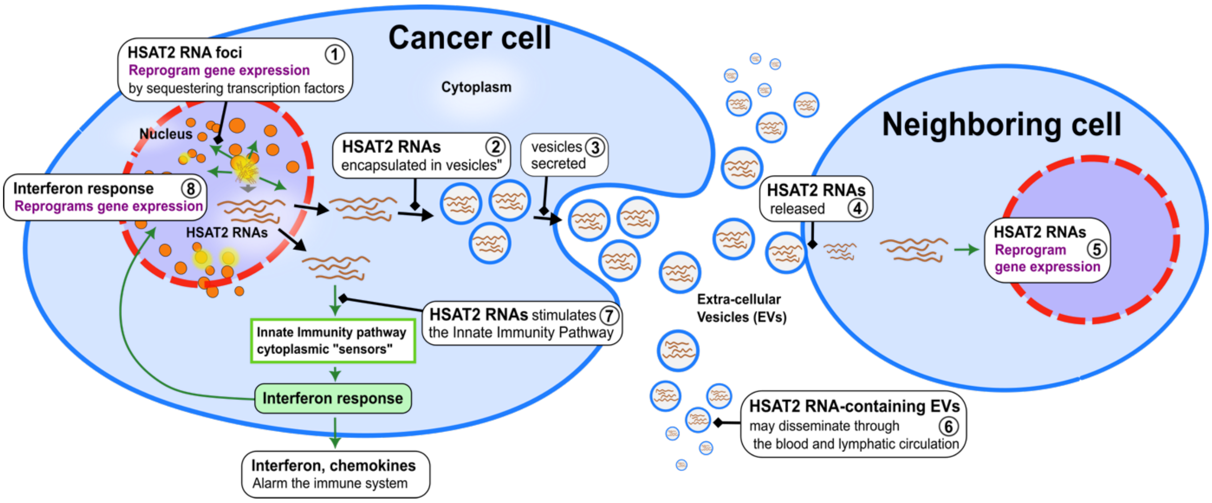

Figure 4. Schematic representation of …